

Ryan Hubbart

(260) 715-9661 | rhubbart@purdue.edu | linkedin.com/in/ryanhubbart | github.com/hub-ry

EDUCATION

Purdue University

Bachelor of Science in Computer Science

Concentrations: Systems Software, Machine Intelligence

Relevant Coursework: Data Structures & Algorithms, Computer Architecture, Object-Oriented Programming, Programming in C, Data Engineering in Python, Linear Algebra, Discrete Math

West Lafayette, IN

Aug. 2024 – May 2028

PROJECTS

Stroke Surgery Clinical Decision Support | *Python, XGBoost, FastAPI, React, Vultr*

March 2026

- Pre-operative tool that predicts safe surgical duration for stroke patients. A problem with no prior data-driven solution. Built as part of a team at a hackathon.
- Two-stage XGBoost pipeline: first stage classifies surgical risk; second stage synthesizes its own regression targets by sweeping duration per patient, since no labeled data exists. Achieved 70% accuracy on held-out labels.
- Engineered 56 features across brain tolerance, oxygen delivery, clot burden, and deterioration rate; 28 derived from clinical literature.
- Used OpenBioLLM-70B to auto-generate pre-op briefs flagging abnormal values; deployed on Vultr with a React frontend.

Event Ticketing System | *Python, FastAPI, SQLite, Docker, Stripe, OpenCV*

Oct. 2025 – March 2026

- Built an end-to-end ticketing system for Purdue Grand Prix (3,000 attendees). QR code generation, Stripe payment integration, and webcam-based gate validation via OpenCV, all offline-capable with local SQLite; replaces third-party ticketing and saves \$1,500 in processing fees.
- Stripe webhook verifies purchase signatures and auto-creates tickets on checkout; gate scanner enforces single-use via DB flag with a 5-second duplicate cooldown.
- Wrote PyTest suite covering ticket integrity, scanning logic, and edge cases; containerized with Docker and pending approval for live deployment at Fall 2026 race.

Self-Hosted Health Data Platform | *Python, FastAPI, Svelte, Docker, Tailscale, Linux*

Sept. 2025 – Feb. 2026

- Designed and deployed a personal health metrics platform on a hardened Linux server. Designed FastAPI backend for time-series ingestion, Svelte dashboard for visualization, containerized with Docker and locked behind a VPN.
- Built the entire stack solo: REST API design, data validation, frontend charting, and firewall-secured remote access.

EXPERIENCE

BoilerMake

Outreach Team Lead

West Lafayette, IN

April 2025 – Present

- Grew Instagram and LinkedIn engagement 1.5× year-over-year for BoilerMake, Purdue's flagship 500-person hackathon, through targeted content and campaign strategy.
- Analyzed demographic data from 1,000+ applications to identify gaps by major, year, and school; redirected outreach toward untapped applicant pools based on the findings.
- Lead a six-person outreach team, coordinating strategy across departments heading into the next cycle.

Purdue Grand Prix Foundation

External Relations Junior Board

West Lafayette, IN

Sept. 2025 – Present

- Coordinated directly with foundation leadership, sponsors, and event organizers to translate operational requirements into a working ticketing and logistics system.
- Managed external operations across ticketing, emergency vehicles, parking, and third-party vendors for a 3,000-person event.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, TypeScript, SQL, R

Web Technologies: HTML, CSS, Node.js, Express.js, React, Svelte, REST APIs

Frameworks & Libraries: FastAPI, Flask, PyTest, Pandas, NumPy, Matplotlib, XGBoost, JUnit

Databases: MySQL, SQLite

Developer Tools: Git, Docker, Linux, Unix, npm, Tailscale